

Erdős Problem #176

EP-176

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $N(k, \ell)$ be the minimal N such that for any $f : \{1, \dots, N\} \rightarrow \{-1, 1\}$ there must exist a k -term arithmetic progression P such that $|\sum_{n \in P} f(n)| \geq \ell$. Find good upper bounds for $N(k, \ell)$. Is it true that for any $c > 0$ there exists some $C > 1$ such that $N(k, ck) \leq C^k$? What about $N(k, 2) \leq C^k$ or $N(k, \sqrt{k}) \leq C^k$?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>